



**Cape Peninsula
University of Technology**

creating futures

**DIPLOMA: INFORMATION AND COMMUNICATION
TECHNOLOGY: APPLICATIONS DEVELOPMENT**

District Six Campus

INDUSTRY GUIDE

2021

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WELCOME

Welcome to the overview of the ICT Diploma offered by the IT Department at CPUT. We hope that you will find the topics interesting and stimulating and that the aspects which will be presented encompass a holistic overview of the programme. The vision of CPUT is to be Africa's leading Smart University of Technology, globally renowned for innovation, with graduates that shape a better world for humanity.

1. PRINCIPLES

Teaching and learning at the Cape Peninsula University of Technology are underpinned by the institutional mission and core values that form the basis of this guide. It is important for students to familiarize them with the mission and core values.

MISSION

Our mission is to develop and sustain an empowering environment where, through teaching, learning, research, and scholarship our students and staff, in partnership with the community and industry, are able to create and apply knowledge in a meaningful way that contributes to development.

CORE VALUES

Integrity, excellence, democracy, respect, accountability, ubuntu, innovation and equity.

2. PURPOSE STATEMENT

This is a career focused ICT qualification that will enable a learner to not only apply the concepts and skills of the specialization in Applications Development sufficiently for a smooth transition into the work place, but also to operate on a higher level in order to solve complex problems adapting to the requirements of the organization's environment.

The purpose of the ICT Diploma in Applications Development is to acquire sufficient knowledge to solve complex problems in an ethical and professional manner by obtaining, processing, evaluating, managing and communicating related information using a range of suitable methods within the area of study as indicated by the different specialization purpose statements.

Although the emphasis is still to acquire, interpret and discuss new knowledge, the **critical evaluation of facts, the practical application** thereof and the **solution of problem situations in practice** are important. In addition to acquiring more **advanced**

skills the emphasis is on developing a person who is **skillful** with a good understanding of his/her **role in society**. The three core components of **knowing, doing and being** are developed for meaningful engagement in the IT discipline and practice on the appropriate level.

Students who complete level 2 are eligible for Work Integrated Learning in which they may be placed in industry during the second semester in their final year of studies for the Diploma in ICT.

3. OVERVIEW

In the **first year** of study, all students enrol for the same subjects. Thereafter, in the **second year** of study a student is required to choose their preferred specialization area.

The Diploma in ICT qualification has three specialisation options, each with its own purpose statement, compulsory and elective subjects. The three specialisation areas are:

- Applications Development
- Communication Networks
- Multimedia Technology

The next three sections in the guide provides an overview of the subjects offered at each year level.

4. DIPLOMA: ICT - FIRST YEAR SUBJECTS

The following subjects are offered in the first year of study in the Diploma: ICT as described in the Table below:

Subject Code	Subject name
Year 1 (Common to ALL Specialisations: APPDEV, COMNET, and MULTIMEDIA)	
ADF152S	Applications Development Foundations 1
BPR152S	Business Practice 1
CNF152S	Communication Networks Foundations 1
ICF152S	ICT Fundamentals 1
MUF152S	Multimedia Foundations 1
PRC152S	Professional Communication 1
PRG152S	Programming 1
PRT152S	Project 1

4.1 APPLICATIONS DEVELOPMENT FOUNDATIONS 1

The purpose of this subject is to provide students with an introduction to applications development using an object-oriented programming language relevant to the Applications Development Industry. Students will learn how to write programs using control structures and procedural programming techniques. Furthermore, the students will also be exposed to an introduction to object-oriented programming. The programming language used as a tool in teaching the curriculum will be Java.

4.2 BUSINESS PRACTICE 1

You will develop a broad knowledge of business operations as well as gaining targeted skills in your specific field of business, such as economics and the world of business, finance, accounting, e-commerce, information and decision making and entrepreneurship.

4.3 COMMUNICATION NETWORKS FOUNDATIONS 1

The focus of Communication Networks Foundations 1 is to provide the knowledge and skills required to design and implement basic home and small business networks. After successfully completing this subject, you will be able to Identify how end-user devices and local networks interact with the global Internet. Explain the requirements for network

connectivity. Build a small network using a virtual network simulator Packet tracer. Explain the importance of IP addressing and explain how the protocols of the TCP/IP suite enable network communication. Demonstrate an understanding of physical, data security and encryption. Troubleshoot common network issues found in home and small business networks.

4.4 ICT FUNDAMENTALS 1

On completion of this subject, the student will develop computer assembling, operating systems installation, and troubleshooting skills as the basis of good computer administration. Students will also be exposed in working with the virtual network simulator (Packet tracer) to design, implement and test the basic network. Given a computer and mobile devices, students will be able to identify the core concepts of computer hardware and software. Prepare Hardware for Operating System installations and understand Operating systems. Students should be able to apply proper safety measures and use preventative measures for both hardware and software. Students should be able to understand different numbering systems and the basic logic operations. Understand cloud computing and the different types of networks and topologies.

4.5 MULTIMEDIA FOUNDATIONS 1

This subject introduces the various aspects of the creative technologies field and industry. The course aims to inform the student on theoretical aspects as well as the practical implementation of each component and provide students with the ability to create.

4.6 PROFESSIONAL COMMUNICATION 1

Professional Communication 1 is a mandatory first year course in the Department of Information Technology. It is designed to equip learners with critical academic and professional skills which can enhance their performance at the university and in the workplace.

4.7 PROGRAMMING 1

The purpose of this subject is to teach students programming concepts which are applied in many programming languages. Students will be introduced to programming

structure, logic and design. This subject will also introduce students to webpage design using HTML5, CSS3 and JavaScript in the second semester.

4.8 PROJECT 1

The purpose of this subject is to equip students with technical and practical skills of Information Technology (IT) project management. The study seeks to achieve this by integrating knowledge from other subjects in the diploma level. The subject guides the student in understanding the role of information systems in an organization, as well as gain an understanding of project life cycle and to discover new project management tools and techniques. Students will learn how to manage time, scope, cost constraints and be able to communicate workable solutions to project stakeholders.

5. DIPLOMA: ICT - SECOND YEAR SUBJECTS

The following subjects are offered in the second year of study in the Diploma: ICT as described in the Table below:

Subject Code	Subject name
Year 2 (Applications Development Specialisation only)	
ADF262S	Application Development Fundamentals 2
ADP262S	Applications Development Practice 2
ADT262S	Applications Development Theory 2
CNF262S	Communication Networks Fundamentals 2
ICE262S	ICT Electives 2
INM262S	Information Management 2
ITS262S	Information systems 2
MAF262S	Multimedia Applications Fundamentals 2
PRC262S	Professional Communications 2
PRT262S	Project 2

5.1 APPLICATIONS DEVELOPMENT FUNDAMENTALS 2

The purpose of this subject is to understand different data structures used in computer programming. Different sorting and searching algorithms will also be introduced. The Java programming language will be used to implement and compare the different data structure concepts.

5.2 APPLICATIONS DEVELOPMENT PRACTICE 2

Applications Development Practice II will guide students to create GUI based applications in Java using swing, client-server applications and to create applications that interact with databases. Other topics include files and serialization. Students will participate in problem solving exercises to eventually understand and implement object-oriented programming concepts.

5.3 APPLICATIONS DEVELOPMENT THEORY 2

This subject is an introduction to Systems Analysis & Design principles and concepts at a basic level. The theory of Systems Analysis & Design as well as practical applications will be covered, with the main emphasis on the analysis phase. Although the emphasis during this year is to acquire, interpret and discuss new knowledge, the critical evaluation of facts, the practical application thereof and the solution of problem situations in practice will be encouraged through the use of a real life business entity.

5.4 COMMUNICATION NETWORKS FUNDAMENTALS 2

The purpose of this subject is to focus on risks, threats, and vulnerabilities associated with the transformation to a digital world and the Internet of Things (IoT). Individuals, students, educators, businesses, organizations, and governments have changed how they communicate, share personal information and media, and do business. This brings to forth what should be taken heed in these times of broadband communications.

5.5 ICT ELECTIVES 2

There are eight Electives subjects offered in the second year as listed below. Students choose one of the Electives across all 3 specialisation areas. The Electives 2 list may change from year to year to probe into the current and relevant trends in the IT industry. Therefore, these subjects may become part of our core curriculum at a later stage.

5.5.1 Functional Programming

5.5.2 Introduction to Artificial Intelligence

5.5.3 Introduction to Game Design and Development

5.5.4 Linux

5.5.5 Mobile Programming 1

5.5.6 PHP Programming

5.5.7 Python Programming 1

5.5.8 R Programming 1 – Statistics

5.6 INFORMATION MANAGEMENT 2

This offering is designed to expose multi-disciplinary students to the tools, techniques and processes that enable the collection and management of information from one or more sources and the distribution of that information to one or more audiences. The coverage includes a study from the perspective of those who have a stake in, or a right to that information. Management means the organization of, and control over the structure, processing, and delivery of information. In addition, the subject introduces all students into the way data is structured in an information technology environment –the nature and use of databases and the ability to retrieve data from relational databases. The ability to use structured query language to retrieve, manipulate and define data and data structures is presented to the audience of this subject.

5.7 INFORMATION SYSTEMS 2

This is an Introduction to database design and provides you with the fundamental knowledge of designing simple databases by means of entity relationship diagrams for implementation in a relational database environment. This course aims to provide practical, hands-on, engaging activities. Leveraging project-based learning techniques, students will create and work with problems which challenge them to design, implement, and demonstrate a database solution for a business or organization.

5.8 MULTIMEDIA APPLICATIONS FUNDAMENTALS 2

This subject teaches the concepts of UX (user experience) and UI (user interface) design. The students are introduced to a myriad of tools and concepts, both offline and online, that can be used to increase the digital experiences regardless of the students chosen stream within the ICT Department. Wireframing, prototyping, accessibility, universal design principles, etc are some of the techniques and concepts implemented to enhance the students' knowledge on the subject.

5.9 PROFESSIONAL COMMUNICATIONS 2

Professional Communication 2 is a mandatory second year course in the Department of Information Technology. It is designed to equip learners with critical academic and professional skills which can enhance their performance at the university and in the workplace.

5.10 PROJECT 2

The purpose of this subject and that of the group project is for the student to gain some experience in group work. These include sharing tasks, running meetings, making collective decisions, dealing with time and people management, writing reports, and giving presentations.

6. DIPLOMA: ICT - THIRD YEAR SUBJECTS

The following subjects are offered in the third year of study in the Diploma: ICT as described in the Table below:

Subject Code	Subject name
Year 3 (Applications Development Specialisation only)	
ADP372S	Applications Development Practice 3
ADT372S	Applications Development Theory 3
ICE362S	ICT Electives 3
ITS362S	Information systems 3
PFP362S	Professional Practice 3
PRT362S	Project 3

PRM372S	Project Management 3
PRP372S	Project Presentation 3

6.1 APPLICATIONS DEVELOPMENT PRACTICE 3

This subject will equip you with the skills to design and implement software that not only correctly model business' need but also performs at scale using the most advanced software architectures. The strategic development skills will help you make best software design choices and integration decisions for any business requirements, resulting in applications and software models that explicitly reflect its core competencies. The tactical development skills will help you design useful software that accurately model unique operations of businesses. The subject will also equip you with knowledge about the broad options available to deploy solutions in a variety of infrastructures, whether on bare metal or in the cloud. Students are exposed to Domain Driven Design, Testing, Version-control systems, Software Design Principles and Patterns, Frameworks. Although this course could also be delivered using C#, Python, Ruby or even PHP, the Java programming language will be used as a programming vehicle to deliver this course since students were already exposed to object-oriented programming in Java in their first two years of study.

6.2 APPLICATIONS DEVELOPMENT THEORY 3

The need for automation to promote efficiency and effectiveness of human operations in diverse fields of application cannot be over-emphasized. The concept of software engineering enables the design and development of software intensive systems based on sound engineering principles. This subject is designed to provide understanding of some of the modern concepts of software engineering and their application to solving industrial scale problems.

6.3 ICT ELECTIVES 3

There are seven Electives subjects offered in the third year as listed below. Students choose one of the Electives across all 3 specialisation areas. The Electives 3 list may change from year to year to probe into the current and relevant trends in the IT industry. Therefore, these subjects may become part of our core curriculum at a later stage.

6.3.1 Blockchain

6.3.2 Business Analysis 2

6.3.3 Kubernetes

6.3.4 Mobile Programming 2

6.3.5 Python Programming 2

6.3.6 R Programming 2 – Data Science

6.3.7 Web Development Frameworks

6.4 INFORMATION SYSTEMS 3

The subject aims to develop the necessary skills, which will enable the student to manage the development and implementation of a Database successfully. Furthermore, the student learns to Design, Develop, and Implement a database using the Relational database model or the Object-oriented database model. Finally, the student is exposed to Analyze data using data warehousing techniques.

6.5 PROFESSIONAL PRACTICE 3

Professional Practice 3 is a mandatory course at the 3rd year level in the Information Technology Department. It is designed to equip learners with critical academic and professional skills which can enhance their performance at the university and place graduates at a competitive advantage in the workplace after graduation. Professional Practice 3 seeks to achieve the above by focusing on the following 3 topics within the following 2 components: Academics Literacies, and Entrepreneurship.

6.6 PROJECT 3

This subject aims to give the student an experience of working in a team on a large-scale project, solving problems relevant to the Information and Communication Technology (ICT) industry. This subject will require the student to consolidate the application and integration of the techniques, skills and concepts learnt over the years in various subjects. Groups are formed and a supervisor is assigned accordingly. Part of the core values of this subject is get you to appreciate how teams work on large projects, work with other students you do not know very well, learn to document your

work, and apply the technical knowledge in applications development to deploy the programming solution.

6.7 PROJECT MANAGEMENT 3

The purpose of this subject is to understand the theory and practice of project management in mature organisations. The course follows the structure of the PM Body of Knowledge (PMBOK). Students will create and work with problems which challenge them to design, implement, and demonstrate a thorough understanding of project management and key tools and techniques defined in the PMBOK.

6.8 PROJECT PRESENTATION 3

Work Integrated Learning (WIL) in this context refers to learning in a workplace as part of a formally structured programme. It can occur within industry, NGOs and other placements in accordance with the demands of a profession or qualification. A learner will be able to integrate specific skills and knowledge with personal and professional values and critical generic learning skills to demonstrate competence in Project Presentation 3. This competence will allow learners to pull together various pieces of learning within a specified context; to demonstrate their ability to integrate learning and performance; to evaluate both, and to identify support or further learning required. This subject allows the IT department to create innovative spaces in which students are encouraged and guided in practising their newly learned competence.

7. FACILITATOR INFORMATION

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Each subject has both:

- a lecturer(s) who teaches the subject
- a subject coordinator for that subject

and students are referred to the subject guides for this contact information.